CHECKING THE RELATIVE CYCLE CONDITION AFTER A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Compatibilities of the cycle operations, in particular proper pushforward commutes with base change (relative-cycles chapter, Lemma lemma-compatibilities, canonical tag 0H4P). cited at proof line 1070. Reduces the general statement to the DVR case: base change commutes with i_*.
- Specialization maps commute with proper pushforward (relative-cycles chapter, Lemma lemma-specialization-proper-pushforward, canonical tag 0H4M). cited at proof lines 1076-1078. Gives i_{0,*} sp_{X/S}(alpha_eta) = sp_{Y/S}(i_{eta,*} alpha_eta) over the DVR.

Source reference. Original lemma 0H4P: sources/stacks-chow/tags.txt, source lines 21359–21359. Cited at proof line 1070: base change commutes with i_*, reducing to the DVR case.

Source reference. Original lemma 0H4M: sources/stacks-chow/tags.txt, source lines 21357–21357. Cited at proof lines 1076-1078: i_{0,*} sp = sp i_{eta,*} over the DVR.

Lemma 0.2. *Let S be a locally Noetherian scheme. Let $i : X \to Y$ be a closed immersion of schemes locally of finite type over S. Let $r \geq 0$. Let α be a family of r-cycles on fibres of X/S. Then α is a relative r-cycle on X/S if and only if $i_*\alpha$ is a relative r-cycle on Y/S.*

Proof. Since base change commutes with i_* (Lemma 0H4P) it suffices to prove the following: if S is the spectrum of a discrete valuation ring with generic point η and closed point 0, then $sp_{X/S}(\alpha_\eta) = \alpha_0$ if and only if $sp_{Y/S}(i_{\eta,*}\alpha_\eta) = i_{0,*}\alpha_0$. This is true because $i_{0,*} : Z_r(X_0) \to Z_r(Y_0)$ is injective and because $i_{0,*}sp_{X/S}(\alpha_\eta) = sp_{Y/S}(i_{\eta,*}\alpha_\eta)$ by Lemma 0H4M. □

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